

C659

Conceptual Physics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A hockey puck made by a company in Ontario slides across smooth ice at a steady 4 m/s in a straight line. Friction is negligible. The rink was resurfaced an hour ago. What is the net force on the puck?

- A. It depends on the puck's mass.
- B. Zero
- C. A forward force that keeps it moving
- D. A backward force equal to its speed

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A 1,200 kg car with a fresh coat of red paint has an engine force of 3,000 N forward and 600 N of friction and air drag backward. The car is two years old. What is its acceleration?

- A. 2,400 m/s²
- B. 2.5 m/s²
- C. 2 m/s²
- D. 3 m/s²

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A net force of 50 N gives a cart an acceleration of 2.5 m/s². What is the cart's mass?

- A. 50 kg
- B. 0.05 kg
- C. 20 kg
- D. 125 kg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A 10,000 kg delivery truck traveling at 15 m/s collides with a 1,000 kg parked car. The truck company has a strong safety record. During the collision, how does the force of the truck on the car compare with the force of the car on the truck?

- A.** They cancel, so neither vehicle accelerates.
- B.** They are equal in size and opposite in direction.
- C.** The truck pushes 10 times harder because it is 10 times heavier.
- D.** The car pushes harder because it is hit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In a science demonstration held in a gym with a 12-meter ceiling, a teacher drops a 7 kg bowling ball and a 0.06 kg tennis ball from the same height at the same time. Air resistance is negligible for this short drop. Which lands first?

- A.** The bowling ball, by a factor of about 117.
- B.** The bowling ball, because it is heavier.
- C.** The tennis ball, because it is lighter.
- D.** They land at the same time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A 500 kg roller coaster car at a park that opened in 1976 starts from rest at the top of a 20 m hill. Friction is negligible and $g = 10 \text{ m/s}^2$. What is its speed at the bottom?

- A.** 400 m/s
- B.** 20 m/s
- C.** About 14 m/s
- D.** 100,000 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A sled on a frictionless slope reaches 10 m/s at the bottom after starting from rest. Using $g = 10 \text{ m/s}^2$, how tall was the slope?

- A. 5 m
- B. 2,000 m
- C. 1 m
- D. 10 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher who brings coffee to class every morning compares a 250 mL cup of 90°C coffee with a 150-liter bathtub of 40°C water. Which statement is correct?

- A. The coffee has the higher temperature, but the bathtub has more thermal energy.
- B. If the coffee were poured into the tub, heat would flow from the tub into the coffee.
- C. They have equal thermal energy because both are liquid water.
- D. The coffee has more thermal energy because it is hotter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A band plays in an outdoor park on a 25°C evening, where sound travels at 340 m/s. The bass note has a frequency of 170 Hz. About 2,000 people attend. What is the wavelength of the note?

- A. About 11.8 m
- B. 57,800 m
- C. 2 m
- D. 0.5 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A sound wave in air travels at 340 m/s and has a wavelength of 0.5 m. What is its frequency?

- A. 680 Hz
- B. 1/680 s
- C. 340 Hz
- D. 170 Hz

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student kit, sold in packs of 30, connects two identical $6\ \Omega$ bulbs in parallel across a 12 V battery. The wires are color-coded red and black. What total current does the battery supply?

- A.** 1 A
- B.** 2 A
- C.** 0.25 A
- D.** 4 A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A third identical bulb is added in parallel to two bulbs already in parallel on a battery. What happens to the circuit's total resistance?

- A.** It increases.
- B.** The other bulbs dim.
- C.** It stays the same, because the bulbs are identical.
- D.** It decreases.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A hockey puck ~~made by a company in Ontario~~ slides across smooth ice at a steady 4 m/s in a straight line. Friction is negligible. ~~The rink was resurfaced an hour ago.~~ What is the net force on the puck?

Target word: net force during constant velocity **Decides:** Steady speed in a straight line means zero acceleration.

A	Dropped element	With zero acceleration, net force is zero for any mass.
B	Correct	Constant velocity means no net force; inertia keeps it moving.
C	Confusion set	The pre-Newton idea that motion needs a force.
D	Adjacent category	Mixes force with velocity; the stem says friction is negligible.

2. Answer C Concept 2

A 1,200 kg car ~~with a fresh coat of red paint~~ has an engine force of 3,000 N forward and 600 N of friction and air drag backward. ~~The car is two years old.~~ What is its acceleration?

Target word: acceleration **Decides:** Net force = 3,000 – 600 = 2,400 N.

A	Adjacent category	That is the net force in newtons, not an acceleration.
B	Dropped element	Uses the engine force alone and ignores friction.
C	Correct	2,400 N / 1,200 kg = 2 m/s ² .
D	Inverted operation	Adds the friction force instead of subtracting it.

3. Answer C Concept 2 · transfer

A net force of 50 N gives a cart an acceleration of 2.5 m/s². What is the cart's mass?

Target word: solve for mass **Decides:** $m = F / a$.

A	Wrong stem number	The force value, not the mass.
B	Inverted operation	Divides acceleration by force.
C	Correct	50 / 2.5 = 20 kg.
D	Inverted operation	Multiplies force by acceleration.

4. Answer B Concept 3

A 10,000 kg delivery truck ~~traveling at 15 m/s~~ collides with a 1,000 kg parked car. ~~The truck company has a strong safety record.~~ During the collision, how does the force of the truck on the car compare with the force of the car on the truck?

Target word: compare the pair of forces **Decides:** Newton's third law pairs are always equal and opposite.

A	Adjacent category	The forces act on different objects, so they do not cancel.
B	Correct	Action-reaction forces are always equal.
C	Confusion set	Mass changes the acceleration each feels, not the force.
D	Inverted operation	Neither pushes harder.

5. Answer D Concept 4

In a science demonstration ~~held in a gym with a 12-meter ceiling~~, a teacher drops a 7 kg bowling ball and a 0.06 kg tennis ball from the same height at the same time. Air resistance is negligible for this short drop. Which lands first?

Target word: which lands first **Decides:** All objects in free fall have the same acceleration.

A	Wrong stem number	Uses the mass ratio (7 / 0.06) as if it set the fall time.
B	Confusion set	The heavier object has more force on it but also more inertia.
C	Inverted operation	Being lighter does not speed a fall.
D	Correct	Both accelerate at g regardless of mass.

6. Answer B Concept 5

A ~~500-kg~~ roller coaster car ~~at a park that opened in 1976~~ starts from rest at the top of a 20 m hill. Friction is negligible and $g = 10 \text{ m/s}^2$. What is its speed at the bottom?

Target word: speed at the bottom **Decides:** $v = \sqrt{2gh} = \sqrt{400}$.

A	Dropped element	Forgets to take the square root.
B	Correct	$\sqrt{2 \times 10 \times 20} = \sqrt{400} = 20 \text{ m/s}$.
C	Dropped element	Leaves out the 2: $\sqrt{10 \times 20}$.
D	Camouflage	Uses the mass to get mgh in joules; mass does not affect the speed.

7. Answer A Concept 5 • transfer

A sled on a frictionless slope reaches 10 m/s at the bottom after starting from rest. Using $g = 10 \text{ m/s}^2$, how tall was the slope?

Target word: solve for height **Decides:** $h = v^2 / (2g)$.

A	Correct	$100 / 20 = 5 \text{ m}$.
B	Inverted operation	Multiplies v^2 by 2g instead of dividing.
C	Confusion set	Uses v instead of v^2 .
D	Dropped element	Divides by g instead of 2g.

8. Answer A Concept 6

A teacher ~~who brings coffee to class every morning~~ compares a 250 mL cup of 90°C coffee with a 150-liter bathtub of 40°C water. Which statement is correct?

Target word: temperature vs. total thermal energy **Decides:** The bathtub has far more water molecules.

A	Correct	Temperature is per-particle; total thermal energy grows with the amount of matter.
B	Inverted operation	Heat flows from higher to lower temperature.
C	Absolute language	Amount and temperature both matter.
D	Confusion set	Treats temperature and thermal energy as the same quantity.

9. Answer C Concept 7

A band plays in an outdoor park on a 25°C evening, where sound travels at 340 m/s. The bass note has a frequency of 170 Hz. About 2,000 people attend. What is the wavelength of the note?

Target word: wavelength **Decides:** $\lambda = v / f$.

A	Camouflage	Divides the attendance by frequency; the crowd size does not matter.
B	Inverted operation	Multiplies speed by frequency.
C	Correct	$340 / 170 = 2$ m.
D	Inverted operation	Divides frequency by speed.

10. Answer A Concept 7 • transfer

A sound wave in air travels at 340 m/s and has a wavelength of 0.5 m. What is its frequency?

Target word: solve for frequency **Decides:** $f = v / \lambda$.

A	Correct	$340 / 0.5 = 680$ Hz.
B	Adjacent category	That is the period, not the frequency.
C	Wrong stem number	The wave speed, not the frequency.
D	Inverted operation	Multiplies speed by wavelength.

11. Answer D Concept 8

A student kit, sold in packs of 30, connects two identical 6 Ω bulbs in parallel across a 12 V battery. The wires are color-coded red and black. What total current does the battery supply?

Target word: total current **Decides:** Two 6 Ω in parallel = 3 Ω .

A	Confusion set	Treats the bulbs as in series (12 Ω).
B	Dropped element	Current through only one bulb.
C	Inverted operation	Divides resistance by voltage.
D	Correct	$12 \text{ V} / 3 \Omega = 4 \text{ A}$.

12. Answer D Concept 8 • transfer

A third identical bulb is added in parallel to two bulbs already in parallel on a battery. What happens to the circuit's total resistance?

Target word: effect of adding a parallel branch **Decides:** Each new branch is another path for current.

A	Confusion set	That is the series rule.
B	Adjacent category	Describes series behavior; parallel bulbs keep full voltage.
C	Absolute language	Identical bulbs still add paths.
D	Correct	More parallel paths lower total resistance.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Inertia (Newton's first law)	With zero net force, an object keeps its velocity; no force is needed to keep something moving.	Believing a moving object must have a force pushing it forward.	Constant speed in a straight line, often on ice or in space.
2	Net force and acceleration	$a = F_{\text{net}} / m$, where F_{net} is the vector sum of all forces.	Using only the applied force and ignoring friction or drag.	An applied force plus an opposing force on the same object.
3	Action-reaction pairs	Two objects exert equal and opposite forces on each other; the forces act on different objects, so they do not cancel.	Thinking the heavier or faster object pushes harder.	A collision or push between two unequal objects.
4	Free fall is independent of mass	Without air resistance, all objects fall with the same acceleration, g .	Expecting heavier objects to fall faster.	Two objects of different mass dropped together, with air resistance negligible.
5	Conservation of mechanical energy	Without friction, mgh at the top becomes $\frac{1}{2}mv^2$ at the bottom, so $v = \sqrt{2gh}$, independent of mass.	Letting mass change the final speed, or forgetting the square root.	A height, a frictionless track or slide, and a speed question.
6	Temperature vs. thermal energy	Temperature measures average kinetic energy per particle; total thermal energy also depends on how much material there is; heat flows from higher to lower temperature.	Assuming the hotter object always holds more thermal energy.	A small hot object compared with a large warm one.
7	Wave speed relation	$v = f\lambda$; frequency is set by the source, and speed is set by the medium.	Multiplying when you should divide, or confusing frequency with period.	Two of speed, frequency, and wavelength given.
8	Series vs. parallel circuits	Adding resistors in parallel lowers total resistance and raises total current; in series it raises total resistance. $I = V / R$.	Assuming more bulbs always means more resistance.	Bulbs or resistors "side by side" or on separate branches.

Course scope (public description): Conceptual Physics provides a broad, conceptual overview of the main principles of physics, including mechanics, thermodynamics, wave motion, modern physics, and electricity and magnetism. Problem-solving activities and laboratory experiments provide students with opportunities to apply these main principles, creating a strong foundation for future studies in physics. There are no prerequisites for this course.